

2 Calibration and characterization of a 3D magnetic 3 Hall sensor card

4 **Lukas Messner^{a,b} Nicola Pacifico^a Daniel Baumgarten^b Raphael Dumps^a**

5 ^a*CERN, EP-DT*

6 *Meyrin, Switzerland*

7 ^b*University of Innsbruck, Biomedical Engineering Group, Department of Mechatronics,*
8 *Innsbruck, Austria*

9 *E-mail:* lukas.messner@cern.ch

10 **ABSTRACT:** High Energy Physics experiments make use of large magnets to identify charged parti-
11 cles through the bending of the particle trajectories. To accurately map the field generated by such
12 magnets, one typically employs 3D magnetic field probes, consisting of 3 individual Hall elements
13 mounted into an orthogonal configuration. In addition to the voltage produced by the Hall effect,
14 single-axis Hall elements are affected by parasitic effects such as offset voltage, temperature depen-
15 dence, nonlinearity, and the planar Hall effect. Mechanical inaccuracies related to their physical
16 integration also affect the final measurement accuracy. These configurations typically thus require
17 a dedicated calibration procedure to compensate for the parasitic effects mentioned above and to
18 enable high-precision magnetic field measurements. This article presents the mathematical model
19 of a previously developed 3D magnetic field sensor card, introduces a novel calibration proce-
20 dure designed to achieve high measurement precision, and characterizes the sensor card for use in
21 high-energy physics experiments for mapping and monitoring applications. The characterization
22 results demonstrate that the Hall card achieves high measurement accuracy, with a mean magnitude
23 error of 9.15×10^{-5} T and a mean angular error of 0.97 mrad over the field range of 0.1 T–2 T.
24 This performance is consistent across temperature variations around room temperature, tested from
25 10 °C to 32 °C. Measurements on a total of 10 sensor cards confirm the reproducibility of both the
26 production process and the calibration procedure.

27 **KEYWORDS:** Calibration and fitting methods, Models and simulations, Data acquisition concepts,
28 Control systems

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44 1 Introduction

45 Precise knowledge of magnetic fields is essential for momentum reconstruction and particle identi-
46 fication in modern high-energy physics experiments. Large solenoidal, toroidal, or dipole magnets
47 installed in particle detectors generate complex three-dimensional field distributions that must be
48 accurately characterized to ensure correct particle identification [1]. While numerical simulations
49 provide initial estimates, detailed field mapping is required to validate and refine these models.
50 Such measurements require compact, high-precision magnetic sensors capable of simultaneously
51 measuring all three components of the magnetic field vector. Three-axis Hall sensors, when prop-
52 erly calibrated, offer a practical and robust solution for this task and are therefore widely employed
53 in the magnetic field mapping systems used for large detector magnets at CERN and other research
54 facilities [2–4].

55 Single-axis Hall elements can achieve very high accuracy, typically in the order of 0.01 % of
56 the measurement range. However, they are affected by several parasitic effects, such as offset voltage
57 in the absence of a magnetic field, temperature-dependent sensitivity, Hall voltage nonlinearity, and
58 the planar Hall effect. For 3D magnetic field measurements, integrated Hall sensors fabricated using
59 CMOS technology can be employed. In these devices, the offset voltage and planar Hall effect are
60 mitigated using the spinning current technique [5]. Nevertheless, their accuracy is generally lower,
61 typically between 0.1 % and 1 %, due to the use of less efficient vertical Hall element geometries,

which makes them less suitable for high-precision field mapping applications. An alternative approach is to combine three orthogonally arranged single-axis Hall elements, which preserves the precision and sensitivity of each element but introduces challenges related to maintaining accurate orthogonality between them. To compensate for all these parasitic effects, 3D Hall sensors require a sophisticated calibration procedure to achieve high-precision measurements [6, 7]. Examples of calibration setups can be found in [8, 9], achieving accuracies in the order of 10^{-4} T. However, calibration procedures are typically tailored to the specific Hall voltage readout system and must be adapted accordingly.

In previous work, the hardware, firmware, and a preliminary calibration of a new 3D magnetic Hall sensor card were presented [10]. To measure the three magnetic field components, three Hall elements are arranged in a three-dimensional Cartesian configuration. This is achieved by assembling a 3D structure from custom one-dimensional support PCBs (Printed Circuit Boards). Once assembled, these support PCBs are inserted into milled slots in the main PCB, where they are mechanically secured and electrically connected through solder joints. A picture of the developed board and the Hall element assembly is shown in Figure 1.

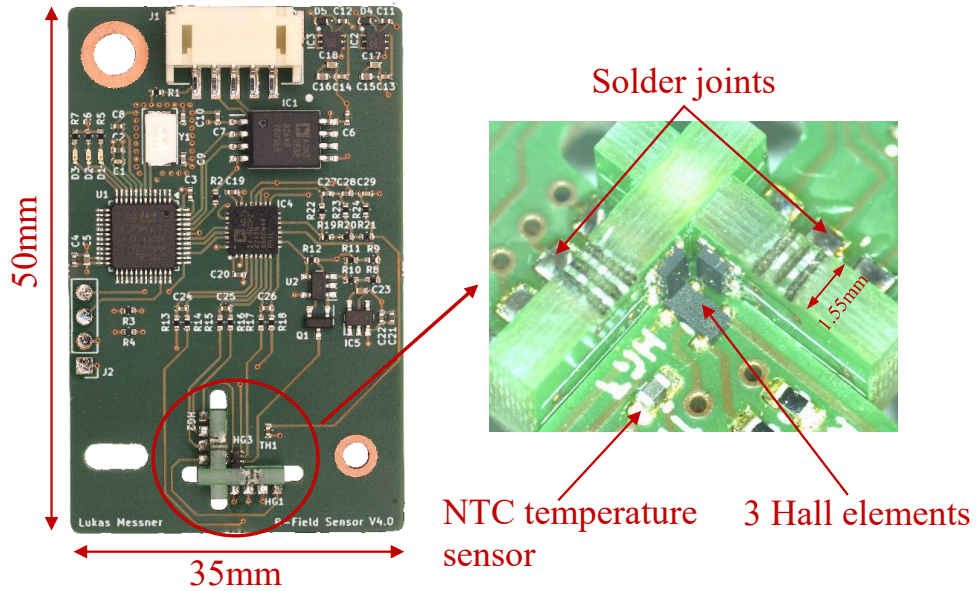


Figure 1. PCB of the previously developed 3D Hall sensor card and 3D placement of the Hall elements with custom one-dimensional support PCBs.

The objective of this paper is to introduce the mathematical model used to describe the Hall voltages of each Hall element in the previously developed 3D Hall sensor card, and to present the newly developed calibration setup. Compared to the setup described in [10], the new system enables calibration up to 2 T and accounts for temperature dependence. The calibration procedure determines a complete set of model parameters while compensating for all parasitic effects described above. It can be performed either at a fixed field magnitude or across a range of field strengths, which in this work spans from 0 T to 2 T. While CERN’s PT-7 dipole magnet can reach fields up to

2.5 T, operation beyond 2 T requires substantially higher coil currents due to magnetic saturation, resulting in increased thermal load. This leads to temperature fluctuations that degrade measurement reproducibility. Limiting the field range to 2 T ensures stable conditions and reliable calibration data.

Furthermore, this article describes the methods used to characterize the sensor card and presents results assessing the performance of the calibration procedure. To meet the accuracy requirements up to 2 T, the sensor card should achieve a field magnitude accuracy in the order of 10^{-4} T [11] and an angular accuracy of approximately 1 mrad [12], consistent with the demands of high-energy physics applications. In addition, the system must maintain this accuracy over temperature variations around room temperature, as experimental caverns, while temperature-controlled, can experience fluctuations during commissioning, operation, and access periods. Finally, the results must be reproducible across different sensor cards and calibration runs to ensure compatibility with medium-scale production of up to 200 units for mapping applications.

2 Mathematical model of the Hall voltages

To model the voltage response of the sensor card, a reference coordinate system must be defined. This coordinate system is placed at the intersection point of the axes of the three Hall elements, located at the horizontal center of the card, as shown in figure 2 (left). The x -axis points toward the connector side of the card, parallel to its vertical edge. The y -axis is parallel to the horizontal edge, forming a right-handed coordinate system with the z -axis oriented perpendicular to the surface of the card. The Hall elements are positioned along the coordinate axes: HG1 in the x -direction, HG2 in the $-y$ -direction, and HG3 in the z -direction. They are located 0.75 mm and 1.05 mm away from the intersection point, as shown in figure 2 (right). Their packaging size prevents smaller distances. To reference the orientation of the Hall elements, The top sides are indicated by a thicker border line.

The goal is to describe the voltage response of each Hall element (V_1, V_2, V_3) as a function of the magnetic field vector, defined in the sensor card's reference coordinate system as $\vec{B}_{\text{card}} = (B_x, B_y, B_z)$. For modeling purposes, it is more convenient to express the magnetic field in spherical coordinates, $\vec{B}_{\text{card}} = (r, \theta, \varphi)$, where

$$r = \sqrt{B_x^2 + B_y^2 + B_z^2}, \quad (2.1)$$

$$\theta = \arccos\left(\frac{B_z}{r}\right), \quad (2.2)$$

$$\varphi = \arctan\left(\frac{B_y}{B_x}\right). \quad (2.3)$$

This transformation provides the basis for using spherical harmonics. Spherical harmonics are a set of orthogonal basis functions that allow continuous functions defined on the surface of a sphere to be expressed as linear combinations of these functions. They are particularly suited for scalar functions depending on angular components (θ, φ), as encountered in this model [8, 13].

The orientation of each Hall element is defined by the rotation matrices R_i for $i = 1..3$, as presented in equations 2.4 - 2.6. HG3, serving as the reference element, lies in the xy plane

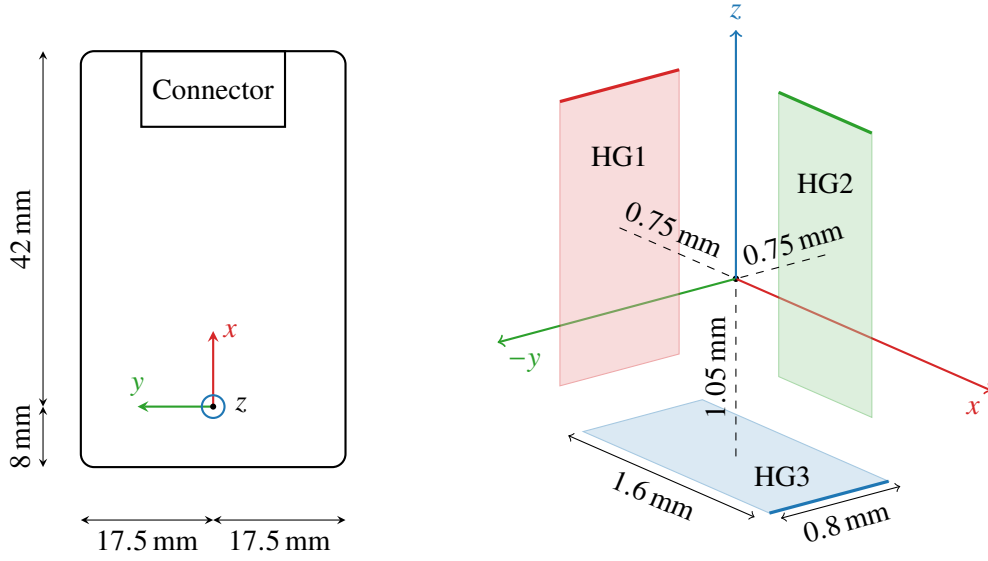


Figure 2. Coordinate system of the 3D Hall sensor card (left) and Hall element placement (right).

with its normal aligned along the z -axis and therefore undergoes no rotation. As illustrated in figure 2, the orientation of HG1 results from a -90° rotation around the y -axis, followed by a 180° rotation around the z -axis, relative to the reference orientation of HG3. Similarly, HG2 is obtained by applying a -90° rotation around the y -axis, followed by a 90° rotation around the z -axis. To account for minor misalignment errors introduced during the assembly and soldering process, each orientation additionally includes small correction angles $\alpha_{\text{corr},i}$ and $\beta_{\text{corr},i}$ around the x and y axes. The rotation of the Hall element around its normal axis, would be required to fully describe its misalignment. However, this degree of freedom is captured by the linear combination of components in the spherical harmonic expansion presented below. Since the objective in the description of the voltage is to rotate the magnetic field vector rather than the Hall element itself, the inverse of the rotation matrices must be applied. The rotated magnetic field vector used in the voltage computation is thus given by $\vec{B}_i = R_i \vec{B}_{\text{card}}$.

$$R_1 = R_{xy}^{-1}(\alpha_{\text{corr},1}, \beta_{\text{corr},1}) R_{zy}^{-1}(180^\circ, -90^\circ) \quad (2.4)$$

$$R_2 = R_{xy}^{-1}(\alpha_{\text{corr},2}, \beta_{\text{corr},2}) R_{zy}^{-1}(90^\circ, -90^\circ) \quad (2.5)$$

$$R_3 = R_{xy}^{-1}(\alpha_{\text{corr},3}, \beta_{\text{corr},3}) \quad (2.6)$$

Using spherical harmonics, the Hall voltages of the three Hall elements are modeled as a series expansion in terms of real spherical harmonic basis functions evaluated at the direction of the magnetic field vector:

$$V_{\text{model},i}(\vec{B}_i, T) = \sum_{l=0}^N \sum_{m=-l}^l r^l K_i^{lm} Y_{lm}^{\text{real}}(\theta, \varphi) \left(1 + K_i^{\text{temp}}(T - T_0)\right) \quad (2.7)$$

where N denotes the order of the spherical harmonic approximation, and K_i^{lm} are the coefficients associated with the corresponding real-valued spherical harmonic functions Y_{lm}^{real} . The real-valued

spherical harmonics are defined as:

$$Y_{lm}^{\text{real}}(\theta, \varphi) = \begin{cases} (-1)^m \sqrt{2} \Re(Y_{lm}(\theta, \varphi)), & \text{if } m > 0 \\ \Re(Y_{lm}(\theta, \varphi)), & \text{if } m = 0 \\ (-1)^m \sqrt{2} \Im(Y_{lm}(\theta, \varphi)), & \text{if } m < 0 \end{cases} \quad (2.8)$$

where Y_{lm} denotes the complex spherical harmonics based on associated Legendre polynomials. For a detailed definition, the reader is referred to [13]. Additionally, a temperature-dependent linear term is introduced through the parameter K_i^{temp} , to account for the slight linear dependence of the output voltage on temperature. Temperature effects are modeled relative to a reference temperature T_0 , such that the calibration coefficients correspond to the sensor response at $T = T_0$. The reference temperature is chosen as the mean temperature of the calibration data in order to minimize parameter correlations and improve numerical stability. This corresponds to room temperature $T_0 = 20^\circ\text{C}$. Evaluating equation 2.7 for the three Hall elements $i = 1..3$, results in a system of three equations dependent on the field vector $\vec{B}_{\text{card}}$ and the temperature T , where each equation contains two unknown angular correction parameters, $\alpha_{\text{corr},i}$ and $\beta_{\text{corr},i}$, one unknown temperature correction parameter K_i^{temp} , as well as $(N+1)^2$ unknown coefficients K_i^{lm} corresponding to the spherical harmonic expansion. These parameters are identified through the calibration procedure described in the following section.

3 Hall card calibration

3.1 Design of the calibration setup

To determine the correction parameters of the model, the sensor card must be exposed to a set of well-defined magnetic field vectors $\vec{B}_{\text{card}}$ under controlled temperature conditions T . The most effective way to apply varying field vectors is to rotate the sensor card to known orientations within CERN's PT-7 dipole magnet that generates a uniform magnetic field between its poles. For this purpose, a custom-built calibration setup (figure 3) was developed, constructed entirely from non-magnetic materials to avoid perturbations to the magnetic field. The setup enables intrinsic rotations around two axes x and z , defining the polar (θ) and azimuthal (φ) angles of the applied magnetic field vector. Both axes are actuated by two piezoelectric motors from Tekceleo: a WLG-75-R motor for the x -axis rotation and a WLG-30-R motor for the z -axis rotation [14]. The actual angular positions are measured using the built-in motor encoder (WLG-30-R) for the γ_z angle, which provides a resolution of 0.045° , and a Netzer VLX-80 absolute rotary encoder [15] for the γ_x angle, which provides a resolution of $4 \times 10^{-5}^\circ$. In the starting position of the calibration, the coordinate system of the Hall sensor card is aligned with that of the calibrator. Figure 3 shows the coordinate system of the Hall sensor card, which is already rotated around the x -axis to improve visibility. The x -axis of the calibrator is coupled to the first motor and is mechanically supported by an aluminum block to bear the weight of the rotating components. The z -axis is directly driven by the second motor. Custom-designed ball bearings, comprising ceramic balls and Teflon cages, are employed to support both rotation axes. The bearing grooves are machined directly into the aluminum structure to ensure compactness and alignment. The calibrator and sensor mount are precisely designed to rotate the card around a fixed point, which corresponds to the origin of the

coordinate system and the intersection point of the Hall element axes. This alignment minimizes the influence of spatial magnetic field inhomogeneities during calibration.

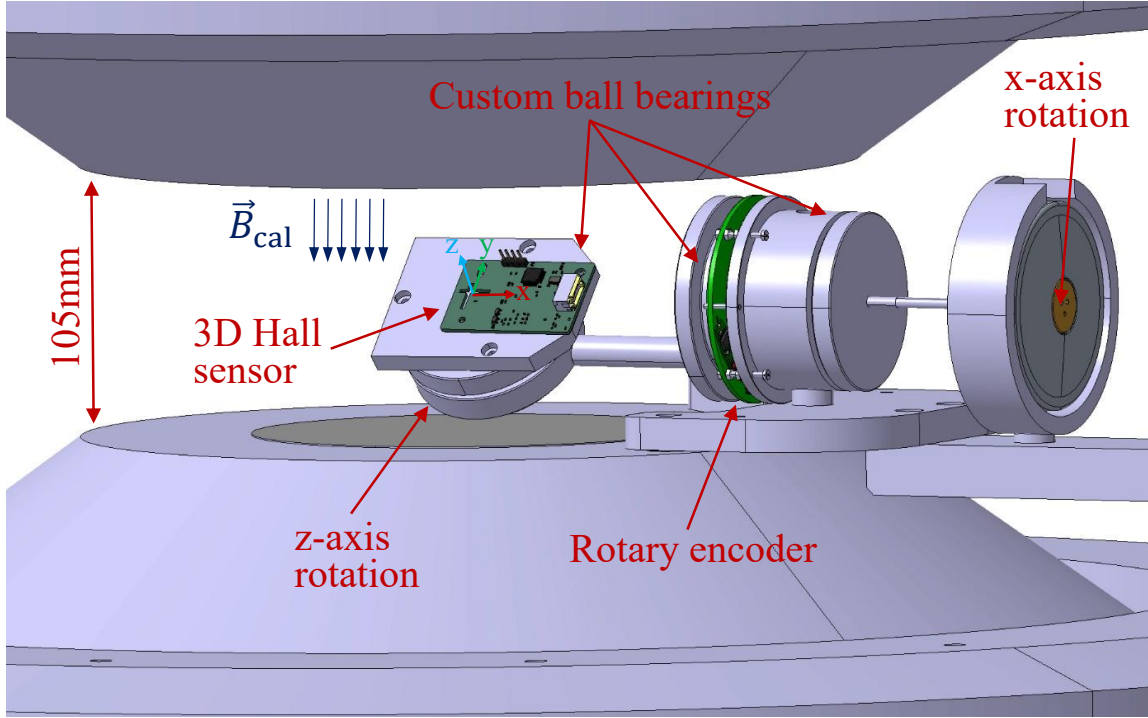


Figure 3. CATIA model of the custom calibration bench inside CERN’s PT-7 dipole magnet.

To maintain a stable temperature T within the calibration environment, a custom-built setup was developed, consisting of a TEC (Thermoelectric Cooling) unit, an automated controller, and a 3D-printed enclosure. The TEC model AA-480-24-44 from Tark Thermal Solutions [16] is a compact cooling system with a cooling power of 480 W. It employs thermoelectric modules to transfer heat between two air streams via forced convection through heat sinks. The unit operates by applying a voltage across Peltier elements, creating a temperature differential that cools one side while heating the other. Compressed air at a pressure of 6 bar is directed through the heat sink on one side, and the tempered air is then supplied to the 3D-printed enclosure surrounding the calibration environment. The PR-59 controller [17] measures the temperature using an NTC thermistor and adjusts the applied voltage, and thus the cooling power, accordingly. The system achieves a temperature stability of $\pm 0.05^\circ\text{C}$, defined as the maximum deviation from the setpoint under steady-state conditions. The mean time per setpoint transition is 14 min, determined from seven consecutive setpoint changes between 14°C and 22°C over a total duration of 95 min. The system can be operated within a temperature range of 10°C to 32°C .

A diagram of the full calibration setup including the temperature regulation can be seen in figure 4. The calibration bench, positioned inside the magnetic field, contains the Hall sensor card, two rotation motors, and a rotary encoder. It is enclosed in a 3D-printed, temperature-regulated housing. The control system, which contains magnetically sensitive components such as the transformers on the motor control boards, is placed outside the magnet. It comprises the power

192 supplies for the control system, encoder, and Hall card; two motor control boards from Tekceleo;
 193 the TEC unit with its controller; and a laptop running a Python script that coordinates the entire
 194 process. The script communicates with the motor control boards via USB to set and read the
 195 motor angles, reads the encoder position via RS-422, sets the desired temperature via RS-232, and
 acquires the Hall voltages from the Hall card via CAN.

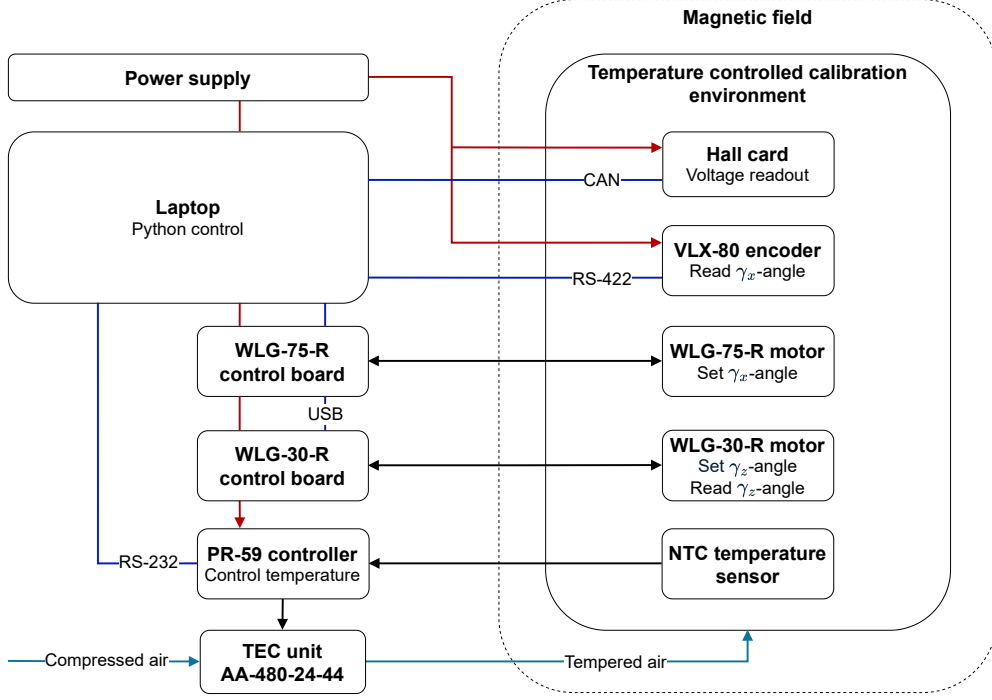


Figure 4. Calibration setup comprising the calibration bench with the Hall sensor card, rotation motors, and encoders inside the magnetic field, and the external control system with temperature regulation located outside due to magnetic sensitivity.

196

197 3.2 Modeling of the calibration setup

198 The applied magnetic field vector in the calibrator's reference frame is constant and defined as
 199 $\vec{B}_{\text{cal}} = (0, 0, -B)$ where B denotes the field magnitude. To use this vector in the Hall voltage model
 200 (equation 2.7), it must be rotated into the reference frame of the sensor card. This transformation is
 201 performed via $\vec{B}_{\text{card}} = R_{\text{cal}} \vec{B}_{\text{cal}}$, where the rotation matrix is given by

$$R_{\text{cal}} = R_{ZXY}^{-1} (\gamma_z, \gamma_x + \gamma_{x,\text{corr}}, \gamma_{y,\text{corr}}). \quad (3.1)$$

202 Here, γ_x and γ_z denote the intended rotations around the respective axes of the calibrator, whereas
 203 $\gamma_{x,\text{corr}}$ accounts for a constant systematic offset in the starting position. A corresponding correction
 204 term for the γ_z rotation was evaluated in the model but could not be reconstructed successfully.
 205 Consequently, the x -axis of the card must be manually aligned as precisely as possible with the
 206 x -axis of the calibrator to ensure accurate results. The term $\gamma_{y,\text{corr}}$ corrects for any tilt of the x -axis
 207 away from the horizontal. The use of capital letters in the subscript of the rotation matrix denotes

intrinsic rotations around these axes, applied in the specified order from right to left. The inverse of the rotation matrix is required to rotate the known field vector $\vec{B}_{\text{cal}}$ into the sensor card's frame of reference. This completes the mathematical model of the calibration system by introducing two additional unknown angular correction parameters into the Hall sensor model described in equation 2.7.

3.3 Calibration procedure

3.3.1 Single-field calibration

The calibration process involves measuring the Hall voltage of each of the three Hall elements under various magnetic field vectors $\vec{B}_{\text{card}}$. This is accomplished by rotating the sensor PCB within a uniform magnetic field of constant magnitude B , while systematically varying the angles θ and φ of the applied field vector. These rotations yield different measured Hall voltages $V_{\text{meas},i}$. Precise knowledge of the field direction is not required, as the calibration reference frame is defined by the rotating plane of the calibration bench. Any misalignments are incorporated into the model as correction parameters. During calibration, the sensor board is rotated within the constant magnetic field and held at a stable temperature of 18 °C. The scan consists of 600 steps, during which the γ_x angle completes five full rotations and the γ_z angle completes six, effectively covering the surface of a sphere. At each step, the γ_x motor advances by 3° and the γ_z motor by 3.6°. At each measurement step, the motors stop, and the voltages from the three Hall elements, along with the corresponding angular positions, and the effective temperature, are recorded. The procedure is then repeated at a second temperature of 23 °C, resulting in a total of 1200 measurements. These temperatures are selected as they lie near the center of the achievable temperature range (10 °C to 32 °C) and can be reached rapidly using the described temperature regulation system without overloading it.

Given the known magnetic field intensity, temperature, and rotation angles γ_x and γ_z , the model described in equation 2.7 becomes a function of both the Hall card and calibration bench parameters. These are grouped into the parameter vector

$$\vec{K} = \left(\alpha_{\text{corr},i}, \beta_{\text{corr},i}, K_i^{\text{temp}}, K_i^{lm}, \gamma_{x,\text{corr}}, \gamma_{y,\text{corr}} \right), \quad (3.2)$$

for $i = 1..3$ with a size of $3(3 + (N + 1)^2) + 2$, where N is the order of the spherical harmonic approximation. The parameters are identified by minimizing the residual function

$$f(\vec{K}) = \sum_{i=1}^{1200} \sum_{j=1}^3 \left(V_{\text{model},i}(\vec{K}) - V_{\text{meas},i} \right)^2 \quad (3.3)$$

for 1200 field configurations applied to the card, using a least-squares optimization approach based on the Trust Region Reflective algorithm. This optimization over the full dataset determines the complete set of model parameters, enabling accurate modeling of the Hall sensor voltage response.

3.3.2 Multi-field calibration

The calibration method described in section 3.3.1 is performed at a specific field strength. However, the resulting parameters vary with field strength and do not scale linearly over the complete calibration range. To ensure accurate sensor performance across a wider field range, calibrations must be conducted at three distinct magnetic field magnitudes: B_1 , B_2 , and B_3 .

243 This yields three sets of parameters $\vec{K}$ corresponding to the different magnetic field magnitudes.
 244 Each parameter K_j of the calibration vector $\vec{K}$ is linearly interpolated between the field magnitudes
 245 according to

$$K_j(B) = \begin{cases} \frac{K_{1j}(B_2 - B) + K_{2j}(B - B_1)}{B_2 - B_1} & \text{if } B_1 \leq B \leq B_2 \\ \frac{K_{2j}(B_3 - B) + K_{3j}(B - B_2)}{B_3 - B_2} & \text{if } B_2 \leq B \leq B_3 \end{cases} \quad (3.4)$$

246 enabling parameter interpolation for any magnetic field value B within the calibrated range.

247 3.4 Magnetic field reconstruction

248 After identifying the Hall card's parameters, the model described in equation 2.7 becomes a system
 249 of three nonlinear equations dependent on the magnetic field vector $\vec{B}_{\text{card}}$ and the temperature T .
 250 Once the three Hall voltages and the temperature near the Hall elements are measured, the magnetic
 251 field vector can be determined by minimizing the following residual function:

$$f(\vec{B}_{\text{card}}) = \sqrt{\sum_{i=1}^3 (V_{\text{model},i}(\vec{B}_{\text{card}}) - V_{\text{meas},i})^2} \quad (3.5)$$

252 using the Nelder–Mead method. Since the Hall card parameters vary with magnetic field strength,
 253 both the parameters and the resulting field vector are computed iteratively, as outlined in algorithm 1.

Algorithm 1 Compute Hall card parameters $\vec{K}$ and magnetic field $\vec{B}_{\text{card}}$

Require: Measured Hall voltages $\vec{V}_{\text{meas}}$, measured temperature T , initial guess $\vec{B}_0$

Ensure: Estimated magnetic field vector $\vec{B}_{\text{card}}$

```

1: while True do
2:   for each  $K_j$  in  $\vec{K}$  do
3:      $K_j \leftarrow K_j(B_0)$  using equation 3.4
4:   end for
5:    $\vec{B}_{\text{card}} \leftarrow$  minimize  $f(\vec{B}_{\text{card}})$  using  $\vec{V}_{\text{meas}}, T, \vec{B}_0$ 
6:   if  $|B_{0,x} - B_{\text{card},x}| \leq 10^{-8}$  and  $|B_{0,y} - B_{\text{card},y}| \leq 10^{-8}$  and  $|B_{0,z} - B_{\text{card},z}| \leq 10^{-8}$  then
7:     return  $\vec{B}_{\text{card}}$ 
8:   else
9:      $\vec{B}_0 \leftarrow \vec{B}_{\text{card}}$ 
10:  end if
11: end while

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254 4 Hall card characterization methods and results

255 4.1 Magnetic field measurement accuracy

256 This measurement investigates if the card's accuracy meets the requirements defined in section 1.
 257 The characterization is designed to determine the card's field range and measurement accuracy. For

this purpose, one card is calibrated as described in section 3.3. CERN’s PT-7 magnet provides the reference field, with an upper limit of 2 T. As outlined in section 3.3.2, the calibration procedure detailed in section 3.3.1 is performed at three different field strengths: $B_1 = 0.1$ T, $B_2 = 1$ T, and $B_3 = 2$ T, at temperatures of 18 °C and 23 °C. The field magnitude is measured using Metrolab’s NMR (Nuclear Magnetic Resonance) teslameter, which has an accuracy of ± 5 ppm [18], and the temperature is regulated by the control system described in section 3.1. The calibration parameters obtained at each field magnitude are then interpolated to be able to determine the card’s parameters over the full calibration range. A second calibration is performed at the field strengths $B_1 = 0.1$ T, $B_2 = 0.5$ T, and $B_3 = 1.0$ T to evaluate differences in accuracy across a different magnetic field range.

In total, the complete calibration uses 3600 measurement points corresponding to 1800 different applied field vectors, since each vector is measured at two different temperatures. All measurement points are used to solve for the magnetic field vector as described in section 3.4. Table 1 reports the mean residuals of the magnitude and angular difference α between the applied and reconstructed magnetic field vectors. The results show that the calibrated Hall card meets the accuracy requirements defined in section 1. For the range 0.1 T–1 T, the mean magnitude error is 5.0×10^{-5} T and the mean angular deviation is 1.2 mrad, with the magnitude error well within the target limit of 10^{-4} T and the angular deviation close to the target of 1 mrad. Extending to the full range of 0.1 T–2 T, the magnitude error slightly increases to 9.5×10^{-5} T while the angular error remains at 1.1 mrad, still consistent with the specified precision.

Table 1. Mean residuals of the magnitude and angular difference α between the applied and reconstructed magnetic field vectors for a calibrated Hall card up to 1 T and 2 T.

Field strength	Mean($ \Delta \vec{B} $)	Mean($ \Delta \alpha $)
0.1 T – 1 T	5.0×10^{-5} T	1.2 mrad
0.1 T – 2 T	9.5×10^{-5} T	1.1 mrad

4.2 Temperature stability

To ensure good accuracy across temperature variation around room temperature, the system model incorporates a temperature correction parameter K_i^{temp} , which compensates for the linear temperature dependence of the Hall voltage. As a result, temperature variations should not affect the accuracy of the magnetic field reconstruction.

To evaluate the effectiveness of the temperature correction, the calibrated Hall card up to 2 T described in section 4.1 is used. The magnetic field of the PT-7 calibration magnet is set to a field strength of 1 T, corresponding to the center of the calibration range. A calibration scan consisting of 600 measurements is then performed at the minimum temperature 10 °C and the maximum temperature 32 °C achievable with the temperature regulation system described in section 3.1. The measurement points obtained from these scans are compared with scans acquired at the calibration temperatures of 18 °C and 23 °C. Table 2 reports the mean residuals of the magnetic field magnitude and the angular difference α between the applied and reconstructed magnetic field vectors. The

lowest residuals are observed at the calibration temperatures of 18 °C and 23 °C, with a field magnitude residual of 8.1×10^{-5} T and 8.3×10^{-5} T respectively. At the temperature extremes, the field magnitude residual increases, reaching 1.6×10^{-4} T at 10 °C and 2.8×10^{-4} T at 32 °C, while the angular error remains below 1.5 mrad over the full temperature range. In addition to the linear temperature dependence of the Hall voltage, the sensor datasheet specifies a temperature-dependent offset voltage [19]. Introducing an explicit temperature-dependent offset term in the model did not lead to a measurable improvement, indicating that offset drift is not the dominant limiting factor. Consequently, temperature effects cannot be fully absorbed by the existing correction scheme, leading to small residual errors at the temperature extremes. These effects primarily impact the reconstructed field magnitude, while leaving the angular reconstruction largely unaffected.

Table 2. Mean residuals of the magnetic field magnitude and the angular difference α between the applied and reconstructed magnetic field vectors for a calibrated Hall card up to 2 T at four different temperatures (10 °C, 18 °C, 23 °C, 32 °C) and a field strength of 1 T.

Temperature	Field strength	Mean($ \Delta \vec{B} $)	Mean($ \Delta\alpha $)
10 °C	1 T	1.6×10^{-4} T	1.5 mrad
18 °C	1 T	8.1×10^{-5} T	1.3 mrad
23 °C	1 T	8.3×10^{-5} T	1.2 mrad
32 °C	1 T	2.8×10^{-4} T	1.5 mrad

4.3 Reproducibility

For medium-scale production intended for mapping campaigns, it is important to achieve reproducible measurement accuracy. To evaluate reproducibility, a total of 10 Hall cards are produced, calibrated, and tested over a calibration range of 0.1 T to 2 T, as described in section 4.1. Table 3 reports the mean residuals of the magnetic field magnitude and the angular difference α between the applied and reconstructed magnetic field vectors for the 10 cards. The results demonstrate a high degree of reproducibility among the calibrated Hall cards. For the full calibration range up to 2 T, the mean residual of the magnetic field magnitude lies consistently between 1.1×10^{-4} T and 7.8×10^{-5} T for the evaluated cards. The corresponding angular deviation remains in a narrow band between 1.1 mrad and 0.8 mrad. The average residual is 9.15×10^{-5} T for the field magnitude and 0.97 mrad for the angular deviation. Despite the limited sample size at this stage, the results already indicate a consistent performance level across independently manufactured and calibrated cards.

5 Conclusion

This article presents the mathematical model developed for a custom magnetic field sensor card incorporating a 3-dimensional Hall sensor array, together with a novel calibration scheme enabling high-precision measurements up to 2 T while remaining compatible with medium-scale production.

Table 3. Mean residuals of the magnetic field magnitude and the angular difference α between the applied and reconstructed magnetic field vectors for 10 calibrated Hall cards in the range 0.1 T to 2 T.

Card number	Field strength	Mean($ \Delta \vec{B} $)	Mean($ \Delta\alpha $)
1	0.1 T – 2 T	9.5×10^{-5} T	1.1 mrad
2	0.1 T – 2 T	8.6×10^{-5} T	1.0 mrad
3	0.1 T – 2 T	9.3×10^{-5} T	1.0 mrad
4	0.1 T – 2 T	8.4×10^{-5} T	0.9 mrad
5	0.1 T – 2 T	1.0×10^{-4} T	1.0 mrad
6	0.1 T – 2 T	7.8×10^{-5} T	0.9 mrad
7	0.1 T – 2 T	8.7×10^{-5} T	0.8 mrad
8	0.1 T – 2 T	1.1×10^{-4} T	1.0 mrad
9	0.1 T – 2 T	8.8×10^{-5} T	1.0 mrad
10	0.1 T – 2 T	9.4×10^{-5} T	1.0 mrad

The approach is based on spherical harmonics, a temperature compensation method, and linear interpolation of the calibration coefficients.

The described approach achieves an average residual of 9.15×10^{-5} T for the magnetic field magnitude and 0.97 mrad for the angular deviation across 10 tested units and over a measurement range up to 2 T. The magnitude error and the angular deviation remain well within the target limits of 10^{-4} T and 1 mrad. The maximum variation of the residuals between the average performance and the individual manufactured units is 1.35×10^{-5} T for the magnetic field magnitude and 0.13 mrad for the angular deviation. This variation remains small compared to the average residuals, indicating that the calibration procedure effectively compensates for manufacturing tolerances. Decreasing the measurement range to 1 T achieves a mean magnitude error of 5.0×10^{-5} T and a mean angular deviation of 1.2 mrad. The slight decrease in magnitude error at lower fields indicates that further refinement of the calibration procedure could improve accuracy near 2 T, although many practical applications involve fields below this level. These results confirm that the card accurately reconstructs both the magnitude and direction of the magnetic field, validating the effectiveness of the multi-field calibration and parameter interpolation procedures.

The temperature compensation method successfully compensates for moderate temperature variations and provides reliable magnetic field reconstruction under changing temperatures around room temperature. Within the tested temperature range between 10 °C and 32 °C, the mean residual of the magnetic field magnitude increases by at most 2×10^{-4} T compared to room temperature conditions. This moderate degradation remains in the same order as the reconstruction accuracy achieved over the full magnetic field range up to 2 T, which is 10^{-4} T.

Overall, the measurement results indicate that the proposed calibration and reconstruction approach provides a reproducible measurement accuracy suitable for medium-scale production and high-precision mapping campaigns. Future work will focus on evaluating the card's radiation

hardness, essential for monitoring applications.

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